

SC Transition Labels:

- empty label ε : a silent transition (e.g. when reducing $\text{skip};C$ to C);
- read label (R, x, v) : reading value v from memory location x
- write label (W, x, v) : writing value v to memory location x
- successful RMW label (U, x, v_o, v_n) : updating the value of location x to v_n when the old value of x in memory is v_o
- failed RMW label $(U, x, v_o, \perp)$: a failed CAS instruction where the old value of x in memory, v_o , does not match the expected value.